BASIC CIVIL ENGINEERING

MODULE-II

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Syllabus of BASIC CIVIL ENGINEERING

Module I

Introduction to Civil Engineering: **Various disciplines of Civil engineering, Importance of Civil engineering in infrastructure development of the country, interdisciplinary nature of construction projects.**

Residential Buildings: **NBC Classification, Basic Components of a building: Basic requirement. Planning and Design of buildings: fundamental requirements, selection of sites, Introduction to building design: functional and structural design.**

Foundations: **Classification, Bearing Capacity of Soil and related terms (definition only)**

Module II

Fundamental Properties of Construction Materials: **Physical, mechanical and durability properties.**

Construction materials: **stone, bricks, cement, aggregate, mortar, concrete, timber, steel, non-ferrous metals, paint, plastic, glass, adhesive, tiles, composites(Definition, classification and application),**

Module III

Importance of Transportation, Transportation modes i.e. Highway, railway, airways, water, pipe and conveyor – Basic Characteristics, advantages and disadvantages. Indian road transport system: Types of roads, classification of highway, urban roads: basic requirements and classification. Basic Components of a Road, Rigid and Flexible pavement (comparison only)

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Module-IV

Quantity of water: Sources of water, Per capita demand, drinking water standards, Public Water Supply System: Necessity and Basic lay out. Conventional water treatment process: Screening, Plain Sedimentation, Sedimentation aided with Coagulation, Filtration, and Disinfection (working principles only).

Module-V

Irrigation: Importance of Irrigation, Classification of Irrigation projects, Irrigation system: Types, Field water distribution, Multipurpose river valley projects, Dams: Purpose, types. Layout of canal Irrigation system: components and definitions.

Course Outcomes

- **Able to understand the basics of civil engineering and fundamental aspects of building.**
- **Able to get the brief overview of general aspect of building material.**
- **Able to get brief idea about transportation modes and planning.**
- **Able to get brief idea about drinking water standards and water treatment plant.**
- **Able to get brief idea about irrigation network system.**

Recommended Books for BASIC CIVIL ENGINEERING

Text books (as per syllabus)

- Gopi, Satheesh. Basic civil engineering. Pearson Education India, 2009.
- S.S. Bhavikatti, Basic Civil engineering, New Age, 2010.

Reference books

- M.S. Palanichamy, Basic Civil Engineering,, McGraw Hill
- B.C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, Basic Civil Engineering, Laxmi Publications
- S.Ramamrutham, Basic Civil Engineering, Dhanpat Rai Publishing Company Private Limited-New Delhi
- S.Ramamrutham, Basic Civil Engineering & Engineering Mechanics, Dhanpat Rai Publishing Company Private Limited-New Delhi
- G.K. Hiraskar, Basic Civil Engineering, , Dhanpat Rai Publishing Company Private Limited-New Delhi
- Ramesh Shah, Basic Civil and Environmental Engineering
- C.P. Kaushik, Basic Civil and Environmental Engineering
- S.K. Duggal, Building Materials, New Age Publishers

Introduction: Scientists Vs Engineer

- Engineering is the art of converting knowledge into useful practical applications
- Engineers contributed more to the shaping civilization than any other professions group
- Role of engineer is to develop technological applications to meet practical needs of society
- Scientists aim to invent while engineer strive the inventions effectively to cater to the needs of mankind
- After the scientific principles of fission was established the hard work of creating atomic weapons and useful power plants accomplished by electrical, chemical and mechanical engineers.

Introduction :Definition of Civil Engineering

- **Civil Engineering: the oldest branch of engineering which has been growing right from the stone-age civilization**
- **Aims to provide a comfortable and safe living for the people.**
- **American Society of Civil Engineering defines**

"*civil engineering as the profession in which knowledge of the mathematical and physical sciences gained by study, experience and practice is applied with judgment to develop ways to utilize economically the materials and forces of the nature for the progressive well-being of man*"

Tasks of a Civil Engineer

- **Investigation:** Collection necessary information/data required for planning
- **Surveying:** Preparation of maps and plans to locate various structures of a project on the surface of earth
- **Planning:** preparation of drawing and preliminary estimate
- **Design:** determination of dimensions and other details
- **Execution:** Preparation of schedule for construction activities, floating tenders, finalization of contracts, supervision of construction work, preparation of bills
- **Maintenance**: look after the maintenance of structures as per requirement

Role of Civil Engineers

- **Measure and map the earth's surface**
- **Plan and develop extensions of towns and cities.**
- **Build the suitable structures for the rural and urban areas for various utilities.**
- **Build the tanks and dams to exploit water resources.**
- **Build river navigation and flood control projects.**
- **Build canals and distributaries to take water to agricultural fields.**
- **Purify and supply water to needy areas like houses, schools, offices etc.**
- **Provide and maintain communication systems like roads, railways, harbours and air-ports.**
- **Devise systems for control and efficient flow of traffic.**
- **Provide, build and maintain drainage and waste water disposal system.**
- **Monitor land, water and air pollution, and take measures to control them. Fast growing industrialization has put heavy responsibilities on civil engineers to preserve and protect environment.**

Civil Engineering: Specializations/Areas

- **Structural engineering**
- **Geotechnical engineering**
- **Water resources engineering**
- **Transportation Engineering**
- **Environment engineering**
- **Surveying and remote sensing**
- **Construction engineering and management**

Importance of Civil Engineering

Importance of Civil Engineering can be viewed considering the importance in the application in following :

- Proper planning of cities and extensions areas with self-sufficient in accommodating offices, educational institutions, markets, hospitals, recreational facility and residential accommodations
- Vertical growth is the need of the time considering the shortage of land in urban areas. Usage of New building technology and innovative methods of analysis could resolve these problems.
- Low cost housing is the need of the day to make poor people afford their own houses.
- Civil engineers have to exploit various water resources and ensure water supply to urban areas throughout the year
- Irrigating water to crops throughout the year by building dams and tanks and bring water to houses through pipes, and to fields through canals and distributaries

Importance of Civil Engineering

•Design of appropriate base course thickness, finishing surfaces, cross drainage, design of horizontal and vertical curves and Proper design of intersection of roads

•Construction of culverts, bridges and tunnels became part of road works

•Construction of railway lines and railway station is an important infrastructure activity. Globalization has resulted in need for building airports and harbours also.

•Other important infrastructural activities of civil engineering are controlling air pollution, noise pollution and land pollution, waste management

• Design of foundations, improving soil behavior by ground improvement techniques and study soil-structure interaction

•Application of Remote sensing and GIS prediction in detecting ground water and prediction of floods, droughts and cyclones

•Development of new technology and material for sustainable and speed construction

Basic Areas of civil Engineering : Surveying

Surveying: determination of relative positions of various objects earth with respect to horizontal and vertical axes through a reference point.

- Earlier, the conventional instruments like chain, tape and levelling instruments were used. In this electronic era, modern electronic equipments like electronic distance meters (EDM) and total stations are used, to get more accurate results easily
- Preparing topo maps of talukas, districts, states and countries and showing all important features like rivers, hills, forests, lakes, towns and cities in plan and elevation (by contour lines) also forms part of surveying
- Science of map making

Basic Areas of civil Engineering : construction Engineering / Technology

- Study of properties and uses of basic materials of construction like stone, bricks, tiles, cement, sand, jelly, steel, glass, glazed tiles, plaster of paris, paints and varnishes.
- Study Behaviour of reinforced cement concrete (R.C.C.) and prestressed concrete
- Exploration of flooring materials, bath room fittings keep on appearing in the market
- Analysis and study of different construction methods and techniques to cater need of the hour
- Study of cost effective methods using various materials
- knowledge quantity of materials and man power requirement
- Use of novel materials and methods of sustainable development

Basic Areas of civil Engineering :structural Engineering & EARTHQUAKE ENGINEERING

- Determination of various loads and induced resultant stress and strain, deflection in different types of structures
- Assessing various types of internal stresses in the components of a structure is known as structural analysis
- Determination of suitable size of the structural component is known as structural design. The structures to be designed may be of masonry, R.C.C., prestressed concrete or of steel.
- Structural engineering involves analysis of various structures like buildings, water tanks, chimneys, bridges etc. and designing them using suitable materials like masonry, R.C.C., prestressed concrete or steel. Adopting optimizing techniques for determination of optimum size of members and materials
- **Earthquake Engineering :** Determination earthquake induced stresses, study of behaviour of structures and designing the structure for earthquake forces constitute earthquake engineering branch of civil engineering.

Basic Areas of civil Engineering :Geotechnical and foundation engineering, Quantity surveying

Geotechnical and foundation engineering: Determination of characteristics of soil and its bearing capacity and methods to improve soil characteristics within the scope of this subject.

- Involves various studies required regarding soil for the design of pavements, tunnels, earthen dams, canals and earth retaining structures along with ground improvement techniques
- Satiability analysis along with design of safe foundation for various infrastructures are within the preview foundation engineering

Quantity surveying:

- Deals with estimation of the quantity of various materials required for a project work
- Deduction for quantity of plastering for various types of openings in the wall, calculating area of painting for various types of doors and windows etc. also form important aspect of quantity surveying
- Labour requirement for various activities of construction

Basic Areas of civil Engineering : Fluid mechanics, irrigation engineering, hydraulic structures, hydrology Ground water hydrology

- **Fluid mechanics and hydraulic machines**: Fluid mechanics is that branch of science which deals with the behaviour of the fluids at rest as well as in motion.
- **Irrigation Engineering**: Identification of suitable water resources and construction of water retaining structures for supply of water to crops by constructing canals, distributaries , aqueducts and regulators form part of irrigation engineering.
- **Hydraulic structures**: Deals with design and construction of hydraulic structures such as dams, water tanks, weir, barrage, shyphon etc.
- **Hydrology**: deals with occurrence and circulation of water. Prediction of floods , draughts and its mitigation techniques
- **Ground water Hydrology**: occurrence, distribution and movement of water under the surface of earth

Basic Areas of civil Engineering : Transportation engineering & Environmental Engineering

Transportation Engineering

Deals with design, construction and maintenance of three modes of transport—land transport, air transport and water transport. Rail transport and road transport are the two components of land transport.

- Highway Engineering: This deals with design of base courses, surface finishes, cross drainage works, road intersections, culverts, bridges and tunnels. Roads need suitable design of horizontal and vertical curves.
- Railway Engineering: This deals with Design construction and maintenance of railway lines and signal systems
- **Airport Engineering:** Design, construction and maintenance of airports and other things related to airlines
- **Docks, harbors and tunnel engineering:** construction and maintenance of Docks, harbors and tunnel Engineering

Environmental Engineering

- Supplying potable water to rural areas, towns and cities and disposal of waste water and solid waste. Solid waste management and disposal of electronic waste
- Study of sources, causes, effects and remedial measures associated with air pollution, water pollution, land pollution and noise pollution

Basic Areas of civil Engineering :Town planning, Infrastructural development, Project management and Remote sensing

- **Town planning:** planning of new cities, extension areas with suitable communication system, educational facilities medical facilities, shopping centres are provided along with residential areas
- **Infrastructural development:** deals with interdisciplinary approach for infrastructure development with collaboration with other branches of engineering such that a newly developed area should have proper approach roads, electricity and water supply, telecommunication facility.
- **Project management:** Planning, scheduling and management forms the basis for taking up a project.
- **Remote sensing:** Remote sensing is science and art of acquiring information about an object without physical contact with it. Remote sensing can be applied for Resource exploration, Environmental study, Land use identification and For assessing and predicting natural hazards.

Early constructions and developments over time

- History of construction is as old as the history of humanity
- Experts agree that the earliest evidence of large-scale buildings is in Mesopotamia. In addition to dwellings, the Mesopotamians built palaces, temples, and ziggurats, often using advanced bricklaying techniques
- In the cities of Ur and Babylon, archaeologists uncovered paved roads dating back to 4000 B.C., which were primarily used for trade.
- The pyramids of Egypt, the temples of Greece, and the imperial palaces of China
- The rise of the construction profession coincided with the rise of modern science in the 17th and 18th centuries.
- Scientific breakthroughs enabled architects and engineers to experiment with a wider array of materials and forms

Development of various materials of construction and methods of construction

- Early building materials were perishable, such as leaves, branches, and animal hides.
- Later, more durable natural materials such as clay, stone, and timber, and, finally, synthetic materials, such as brick, concrete, metals, and plastics were used.
- In old Stone Age (9000 BC to 5000 BC), tools available were made from natural materials including bone, antler, hide, stone, wood, grasses, animal fibers, and the use of water
- Copper Age and Bronze Age construction: Bronze is made when tin is added to copper and brass is made copper with zinc. Copper came into use before 5,000 BC and bronze around 3,100 BC; Mohenjo-daro & Harappa: Bronze, brick

Iron Age construction (1200 BC-50BC): Construction in ancient Mesopotamia:The chief building material was the mud-brick, formed in wooden moulds. Construction in ancient Egypt: stone

- The great Roman development in building materials was the use of hydraulic lime mortar called Roman cement.
- In eighteen century, Portland cement was produced.
- In last two centuries, steel, concrete, composites etc. have been developed

Different construction materials

Various materials are used for constructing buildings, bridges, roads, retaining walls and dams

•Stones	**•Bricks**	**•Sand**	**•Reinforcing steel**
•Cement	**•Plain cement concrete (PCC)**	**•Reinforced cement concrete**	**•Pre-stressed concrete**
•Precast concrete	**•Recycled materials (recycled aggregates, crusher dust, fly ash, blast furnace sand, red mud, polythene)**		
•Lime	**•Tiles**	**•Timber**	**•Aluminum**
•Cast iron	**•Glass**	**•Rubber**	**•PVC**
•Plaster of Paris	**•Asbestos**	**•Paints**	**•Adhesives**
•Varnish	**•Composite materials**	**•Alloy**	**•Polymers**

Stones : Introduction and Classification

- Available in the form of rocks, which is cut to required size and shape and used as building block.
- Stones used for civil engineering works may be classified in three ways: Geological, Physical and Chemical

Geological Classification

Based on their origin of formation stones are classified into three main groups—Igneous, sedimentary and metamorphic rocks.

Igneous rocks : Rocks formed by cooling and solidifying of the magma (the molten or pasty material) is known as igneous rocks. The igneous rocks may be of :

1. **Plutonic rocks**: formed due to cooling of magma considerable depth from earth's surface. The cooling is slow and rocks possess crystalline structure. Ex: Granite
2. Hypabyssal rocks: formed due to cooing of magma at a relatively shallow depth. The cooling is quick and finely grained crystalline structure. Ex. Dolerite

Stones : Geological Classification

Volcanic rocks: formed due to pouring of magma at earth's surface. Cooling is very rapid & rocks are extremely fine grained. Ex. Basalt.

Sedimentary rocks: rocks are formed due to the deposition of products of weathering on pre-existing rocks. The products are carried away from their place of origin by agents like rain, frost, wind, flowing water.

- **Residual deposit:** These are the products of weathering remain at the place of origin
- **Sedimentary deposits:** The insoluble products of weathering are carried away in suspension and when such products are deposited are called sedimentary deposits.
- **Chemical deposits**: materials which are carried away in solution may be deposited by some physio-chemical processes such as evaporation, precipitation etc. are give rise to chemical deposits
- **Organic deposits :** these are formed through the agency of organisms.

Examples of sedimentary rock: gravel, sandstone, lime stone, gypsum,

Stones : Geological Classification

Metamorphic rocks: formed by change in character of pre-existing rocks. The igneous and sedimentary rocks change their character when subjected to great heat and pressure. This process of change is known as **metamorphism.** Four types of metamorphism occur:

a. Thermal metamorphism: heat is dominant factor

b. Cataclastic metamorphism: pressure is the dominant factor (formed at surface)

c. dynamo-thermal metamorphism: combination of stress and temperature (up to certain depth)

d. Plutonic metamorphism: uniform pressure and heat (great depth)

Example: marble (lime stone), sand stone (quartzite)

Stones : Physical Classification

- **Stratified Rocks:** These rocks are having layered structure. They possess planes of stratification or cleavage. They can be easily split along these planes. Sand stones, lime stones, slate etc. are the examples of this class of stones.
- **Unstratified Rocks:** These rocks are not stratified and possess crystalline and compact grains. They cannot be split in to thin slab. Granite, trap, marble etc. are the examples of this type of rocks.
- **Foliated Rocks:** These rocks have a tendency to split along a definite direction only. The direction need not be parallel to each other as in case of stratified rocks. This type of structure is very common in case of metamorphic rocks

Ex: quartz, fedspar

Stones : chemical Classification

Based upon the chemical composition rocks may be of

- ***Silicious rocks:*** Main content of these rocks is silica. They are hard and durable. Examples of such rocks are granite, trap, sand stones etc.
- ***Argillaceous rocks:*** The main constituent of these rocks is argil i.e., clay. These stones are hard and durable but they are brittle. They cannot withstand shock. Slates and laterites are examples of this type of rocks
- **Calcareous rocks:** The main constituent of these rocks is calcium carbonate. Limestone is a calcareous rock of sedimentary origin while marble is a calcareous rock of metamorphic origin

Stones : uses

- Construction of foundations, walls, columns, arches & flooring.
- Stone slabs are used as damp proof courses, lintels and even as roofing materials.
- Stones with good appearance are used in face works
- Used for constriction of piers and abutments of bridges, dams and retaining walls.
- used for paving of roads, footpaths and open spaces round the buildings
- Crushed stones are used for as railway ballast, as a basic inert material in concrete and For making artificial stones and building blocks
- Crushed stones with graved are used to provide base course for roads. When mixed with tar they form finishing coat

Properties of Stones

- **Structure:** The structure of the stone may be stratified (layered) or unstratified**.** Structured stones can be easily dressed and suitable for super structure. Unstratified stones are hard and difficult to dress and are preferred for the foundation works.
- **Texture**: Fine grained stones with homogeneous distribution look attractive and hence they are used for carving. Such stones are usually strong and durable
- **Density**: Denser stones are stronger. Light weight stones are weak. Hence stones with specific gravity less than 2.4 are considered unsuitable for buildings.
- **Appearance**: A stone with uniform and attractive colour is durable, if grains are compact. Marble and granite get very good appearance, when polished. Hence they are used for face works in buildings
- **Strength**: Strength is an important property to be looked into before selecting stone as building block. Indian standard code recommends, a minimum crushing strength of 3.5 N/mm2 for any building block.

- **Hardness**: It is an important property to be considered when stone is used for flooring and pavement. Coefficient of hardness is to be found by conducting test on standard specimen in Dory's testing machine.
- **Percentage wear**: It is measured by attrition test. It is an important property to be considered in selecting aggregate for road works and railway ballast. A good stone should not show wear of more than 2%
- **Porosity and Absorption**: All stones have pores and hence absorb water. The reaction of water with material of stone cause disintegration. Absorption test is specified as percentage of water absorbed by the stone when it is immersed under water for 24 hours. For a good stone it should be as small as possible and in no case more than 5
- **Weathering**: Rain and wind cause loss of good appearance of stones. Hence stones with good weather resistance should be used for face works.
- **Toughness**: The resistance to impact is called toughness. It is determined by impact test. Stones with toughness index more than 19 are preferred for road works. Toughness index 13 to 19 are considered as medium tough and stones with toughness index less than 13 are poor stones
- **Resistance to Fire:** Sand stones resist fire better. Argillaceous materials, though poor in strength, are good in resisting fire.

- **Ease in Dressing:** Cost of dressing contributes to cost of stone masonry to a great extent. Dressing is easy in stones with lesser strength.
- **Seasoning:** The stones obtained from quarry contain moisture in the pores. The strength of the stone improves if this moisture is removed before using the stone. The process of removing moisture from pores is called seasoning. The best way of seasoning is to allow it to the action of nature for 6 to 12 months.

Requirements of Good Building Stones

- Strength: stone should have adequate strength.
- **Durability**: Stones selected should be capable of resisting adverse effects of natural forces like wind, rain and heat.
- **Hardness:** The stone used in floors and pavements should be able to resist abrasive forces caused by movement of men and materials over them.
- **Toughness:** Building stones should be tough enough to sustain stresses developed due to vibrations.

Specific Gravity: Heavier variety of stones should be used for the construction of dams, retaining walls, docks and harbours.
Porosity and Absorption: Building stone should not be porous. If it is porous rain water enters into the pour and reacts with stone and crumbles it. In higher altitudes, the freezing of water in pores takes place and it results into the disintegration of the stone.
Dressing: Giving required shape to the stone is called dressing. It should be easy to dress so that the cost of dressing is reduced
Appearance: In case of the stones to be used for face works, where appearance is a primary requirement, its colour and ability to receive polish is an important factor.
Seasoning: Good stones should be free from the quarry sap. Laterite stones should not be used for 6 to 12 months after quarrying. hey are allowed to get rid of quarry sap by the action of nature. This process of removing quarry sap is called seasoning.
Cost: Cost is an important consideration in selecting a building material. Proximity of the quarry to building site brings down the cost of transportation and hence the cost of stones comes down.

Bricks: as a construction material

- Bricks are artificially made rectangular blocks
- Clay is molded to form rectangular blocks of standard size, which are dried and later burnt to a high temperature to make them dense and compact

Bricks are important building material owing to :

- Easy availability, good strength and durability, insulating and fire resisting properties
- Bricks don’t need dressing and bricks can be laid in simple process not requiring skilled labor.
- Bricks are used for construction of walls, pillars and in some cases floor and roofs
- Brick masonry can easily constructed
- Bricks are economical
- Brick masonry does not require skilled labor
- Being light weight do not need special lifting devices to lift them

Bricks :as a construction material

- Bricks are more fire resistant than stones
- Bricks have uniform size and can be laid accurately in the bond
- Construction of brick masonry is faster than that of stone masonry
- Brick masonry require less mortar than stone masonry

Uses of Bricks:

- Mainly used for the construction of walls
- Can be used in the drains when moulded in the shapes gutter
- Bricks with cavities known as hollow bricks can used in multistoried framed structure
- Bricks made from fire clay can be used as refractory bricks
- Can be used in construction of compound walls, columns etc.
- Broken pieces of **bricks** are used as aggregates in concrete.
- Bricks of superior quality can be used as facing of the wall
- Contsruction of chimneys and other special works

Bricks :as a construction material

- The size of the bricks are of **190 mm × 90 mm × 90 mm**. **With mortar joints**, the size of these bricks are taken as **200 mm × 100 mm × 100** mm

Types of Bricks as per usage:

1. **Building bricks:** bricks are used for the construction of walls.
2. **Paving Bricks:** These are vitrified bricks and are used as pavers
3. **Fire Bricks:** These bricks are specially made to withstand furnace temperature. Silica bricks belong to this category.
4. **Special Bricks:** Facing bricks, Burnt clay hollow bricks, Sewer bricks

Classification of Bricks Based on their Quality

- **First Class Bricks:** These bricks are of standard shape and size & are burnt in kilns.
- **Second Class Bricks:** These bricks are ground moulded and burnt in kilns. The edges may not be sharp and uniform. Such bricks are commonly used for the construction of walls which are going to be plastered
- **Third Class Bricks:** These bricks are ground moulded and burnt in clamps and edges are somewhat distorted. They produce dull sound when struck together. They are used for temporary and unimportant structures
- **Fourth Class Bricks:** These are the over burnt bricks and dark in colour. The shape is irregular and are used as aggregates for concrete in foundations, floors & roads

Bricks :as a construction material

Constituents of bricks :

- **Alumina : A good brick should have 20%-30% of alumina. It imparts plasticity to the to the brick earth.**
- **Silica: A brick earth should contain 50%-60% of silica. Silica prevents cracking, shrinking, and wrapping of raw bricks. The durability depends upon the proper proportion of silica.**
- **Lime: Up to 5% of lime is desirable in good brick earth. Prevents shrinkage of bricks. Bricks may melt and loose their shape owing to excess amount of silica.**
- **Oxide of iron: Up to 5-6% is desirable in good brick earth. This gives red colour to bricks.**
- **Magnesia : It may be allowed to a small extent. Excess of magnesia causes decay of bricks.**

Harmful ingredients: Excess lime, Iron pyrites, alkalies, pebbles and organic matter and vegetation

Qualities of good Bricks

- Colour: Colour should be uniform and bright.
- **Shape:** Bricks should have plane faces. They should have sharp and true right angled corners
- **Size:** Bricks should be of standard sizes as prescribed by codes.
- **Texture:** Bricks should possess fine, dense and uniform texture. They should not possess fissures, cavities, loose grit and unburnt lime.
- **Soundness:** When struck with hammer or with another brick, it should produce metallic sound.
- **Hardness:** Finger scratching should not produce any impression on the brick.
- **Strength:** Crushing strength of brick should not be less than 3.5 N/mm^2. A field test for strength is that when dropped from a height of 0.9 m to 1.0 mm on a hard ground, the brick should not break into pieces

Qualities of good Bricks

- **Water Absorption:** After immersing the brick in water for 24 hours, water absorption should not be more than 20 per cent by weight. For class-I works this limit is 15 per cent.
- **Efflorescence:** Bricks should not show white patches when soaked in water for 24 hours and then allowed to dry in shade. White patches are due to the presence of sulphate of calcium, magnesium and potassium. They keep the masonry permanently in damp and wet conditions.
- **Thermal Conductivity:** Bricks should have low thermal conductivity, so that buildings built with them are cool in summer and warm in winter.
- **Sound Insulation:** Heavier bricks are poor insulators of sound while light weight and hollow bricks provide good sound insulation.
- **Fire Resistance:** Fire resistance of bricks is usually good. In fact bricks are used to encase steel columns to protect them from fire.